Engineering Units

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Types of Units

- Fundamental units:
  - Mass
  - Length
  - Time
  - Temperature
- Derived units (expressed in terms of fundamental units):
  - Force
  - Pressure
  - ...

For example, the unit for force can be derived from the law defining it.
Force = mass x acceleration
mass has a fundamental unit.
acceleration = velocity / time
velocity = length / time
Combining it together:
unit of Force $=$ unit of Mass $\times$ unit of Length / (unit of Time)^2

Système International (SI) and Imperial system of units

- SI system of units is the standard throughout the world
- Imperial or English system is still used in industry

Fundamental units:

Units	**SI units**	**Imperial units**	**Conversion**
Mass	kilogram (kg)	pound mass (lbm)	1 kg = 2.2046 lbm
Length	meter (m)	foot (ft)	1 m = 3.2808 ft
Time	second (s)	second (s)	–
Temperature	Kelvin or degree Celsius (K or °C)	Rankine or degree Fahrenheit (R or °F)	1 K = 1.8 R

Derived units:

Units	**SI units**		**Imperial units**	
Force	Newton (N)	$1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$	pound force (lbf)	$1\,\mathrm{lbf} \approx 32.2\,\mathrm{lbm\,ft/s^2}$
Pressure	Pascal (Pa)	$1\,\mathrm{Pa} = 1\,\mathrm{N/m^2}$	pounds per square inch (psi)	$1\,\mathrm{psi} = 1\,\mathrm{lbf/in^2}$

Exercise problems

Problem 1: Convert 72 km/hr to m/s.
We know: 1 km = 1000 m; 1 hr = 3600 s

$$\begin{array}{llll} 72 & \text{km} & / & \text{hr} \\ = 72 & 1000\ \text{m} & / & (3600\ \text{s}) \\ = 72 \times 1000\ /\ 3600 & \text{m} & / & \text{s} \\ = 20 & \text{m} & / & \text{s} \end{array}$$

Therefore, $72\,\text{km/hr} = 20\,\text{m/s}$.

Problem 2: Convert 100 psi to Pa.

We know: $1\,\text{psi} = 1\,\text{lbf/in}^2$;
$1\,\text{lbf} \approx 32.2\,\text{lbm ft/s}^2$;$1\,\text{ft} = 12\,\text{in}$;
$1\,\text{kg} = 2.2046\,\text{lbm}$;$1\,\text{m} = 3.2808\,\text{ft}$

$$\begin{array}{llll} 100 & \text{psi} & & \\ = 100 & \text{lbf} & / & \text{in}^2 \\ = 100 & 32.2\,\text{lbm ft/s}^2 & / & (1/12\,\text{ft})^2 \\ = 100 \times 32.2 \times 12^2 & \text{lbm} & / & (\text{ft s}^2) \\ = 463680 & \text{lbm} & / & (\text{ft s}^2) \\ = 463680 & 1/2.2046\,\text{kg} & / & (1/3.2808\,\text{m s}^2) \\ = 463680/2.2046 \times 3.2808 & \text{kg} & / & (\text{m s}^2) \\ = 690031 & \text{N/m}^2 & & \\ = 690031 & \text{Pa} & & \end{array}$$

Therefore, 100 psi is approximately 690,000 Pa.